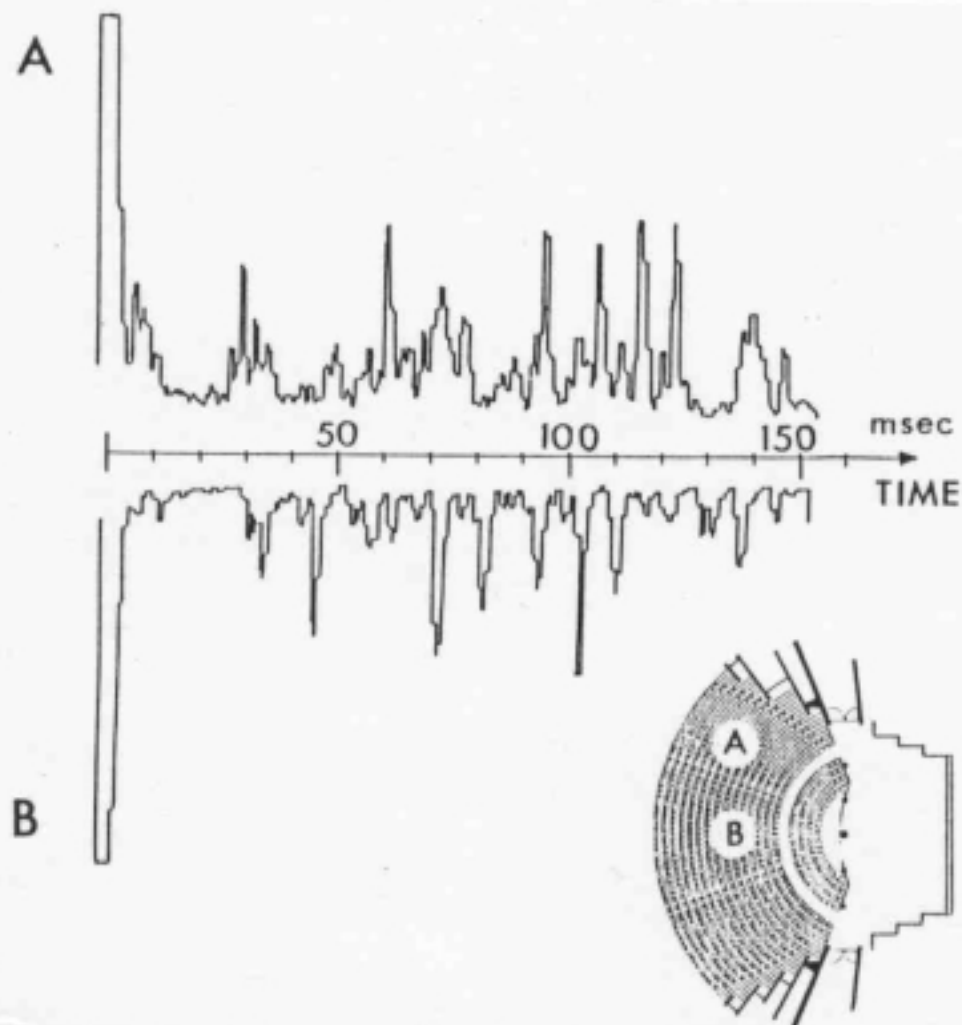




Measured pulse response using pulses from a pulse generator with octave frequency 2000 Hz. The loudspeaker was positioned in the middle of the orchestra enclosure. The finer details in the echograms were investigated using a highly directional microphone. This microphone unit had a parabolic screen with a diameter of 1 m. The first strong reflection seen in the two echograms (time delay 30–35 msec) comes from the ceiling. The next strong reflections (time delay 45–120 msec) come from the side wall reflectors and the freely hanging reflectors below the ceiling. They therefore all have a lateral incidence.

Figure 1. Echogram of the Grieg Memorial Hall in Bergen, Norway (from Strom et al)

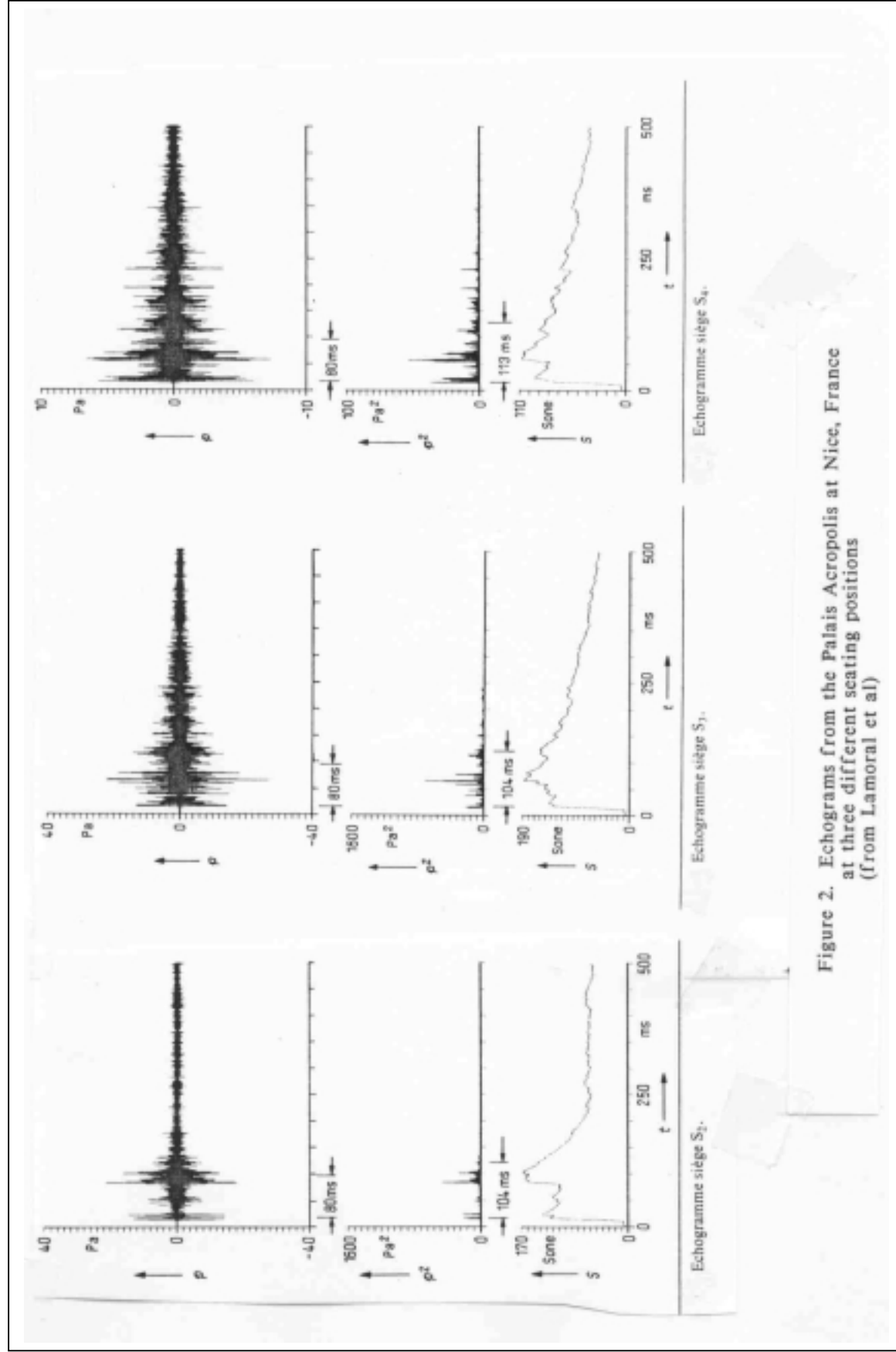


Figure 2. Echograms from the Palais Acropolis at Nice, France
at three different seating positions
(from Lamoral et al)

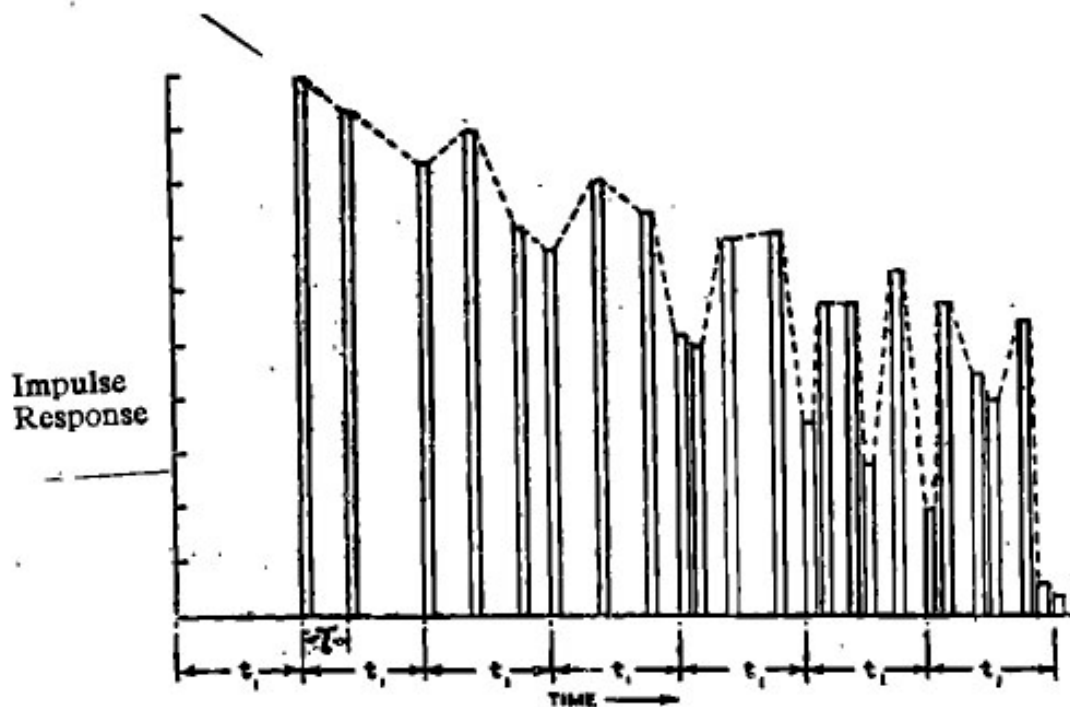
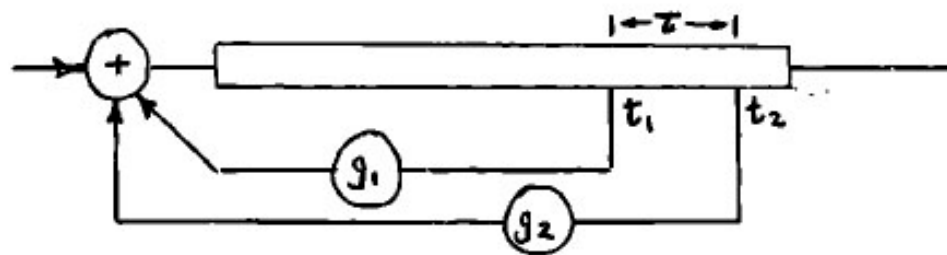
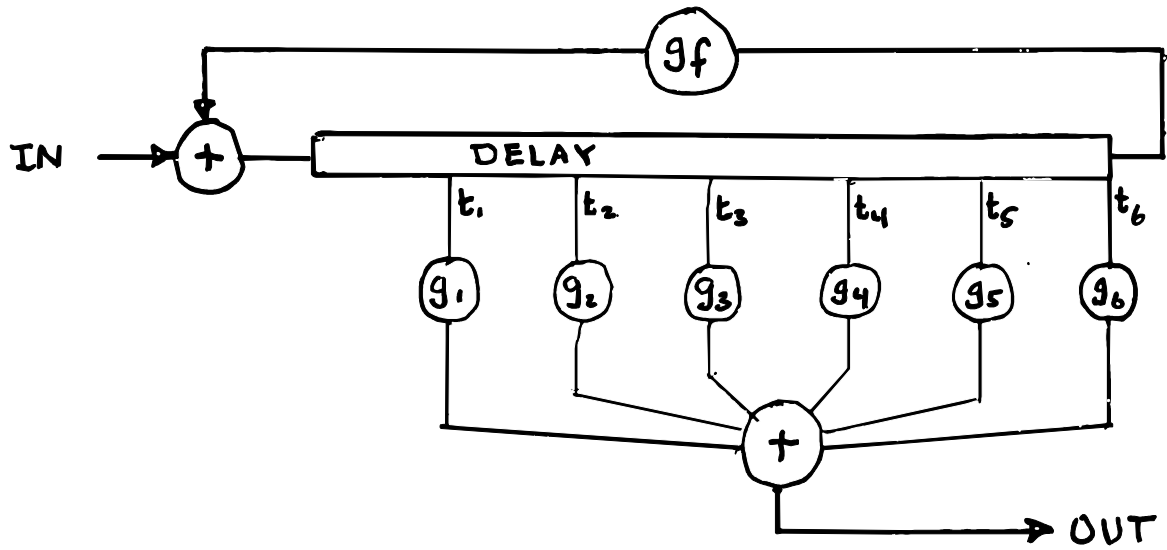


Figure 5. Delay with multiple feedback paths
 Note the density of the echos in the impulse response increases with time, but the amplitudes fluctuate (from Axon et al)



Impulse
Response

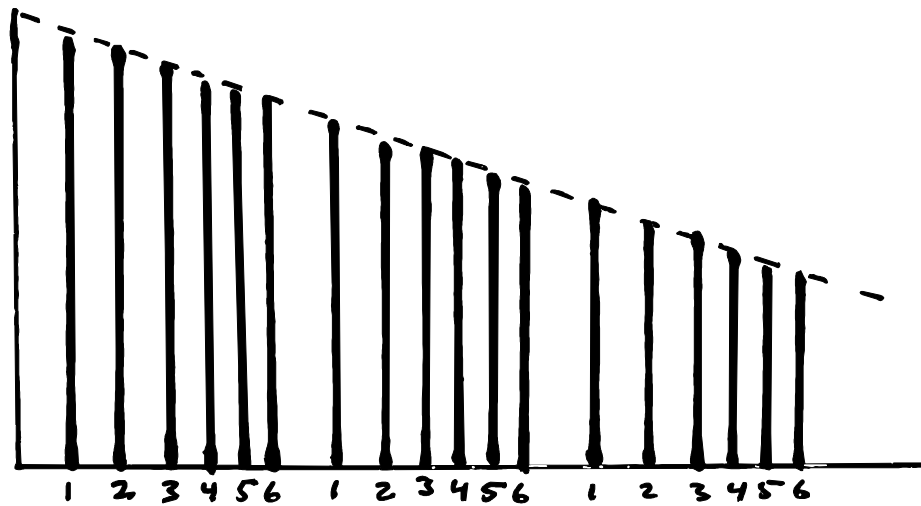


Figure 4. Delay with a single feedback and multiple outputs
If the output gains are chosen correctly, exponential decay results, but with a repeating pattern
(from Axon et al)

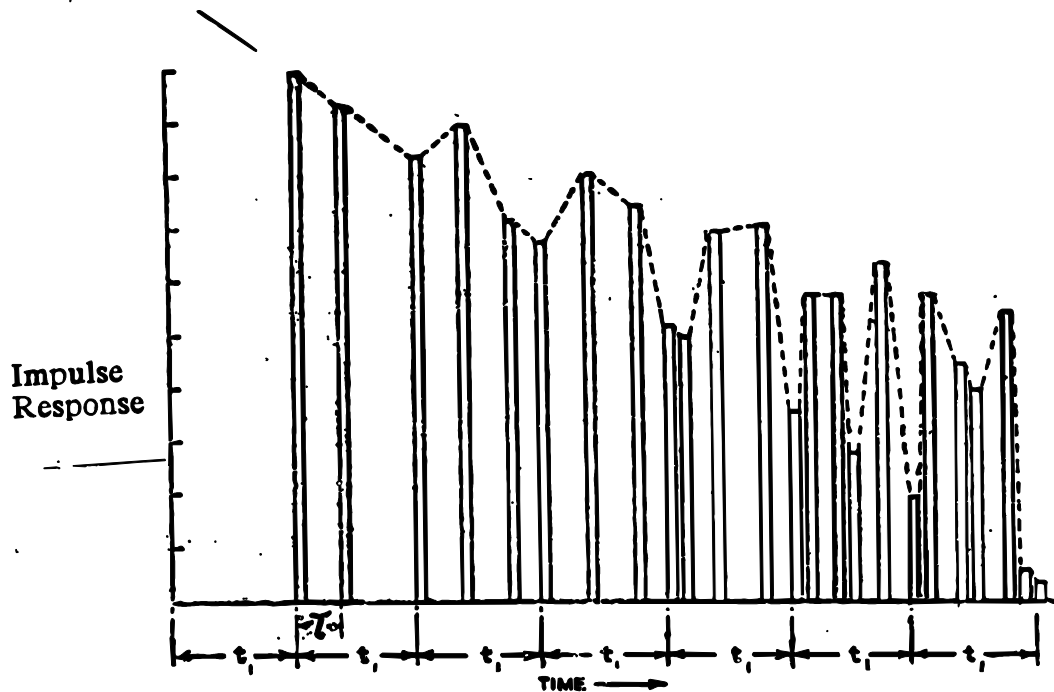
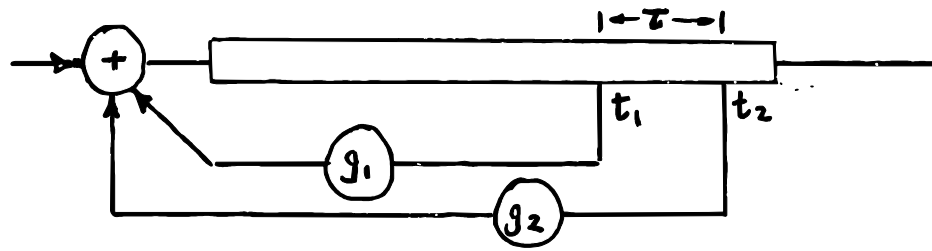


Figure 5. Delay with multiple feedback paths
 Note the density of the echos in the impulse response increases with time, but the amplitudes fluctuate (from Axon et al)

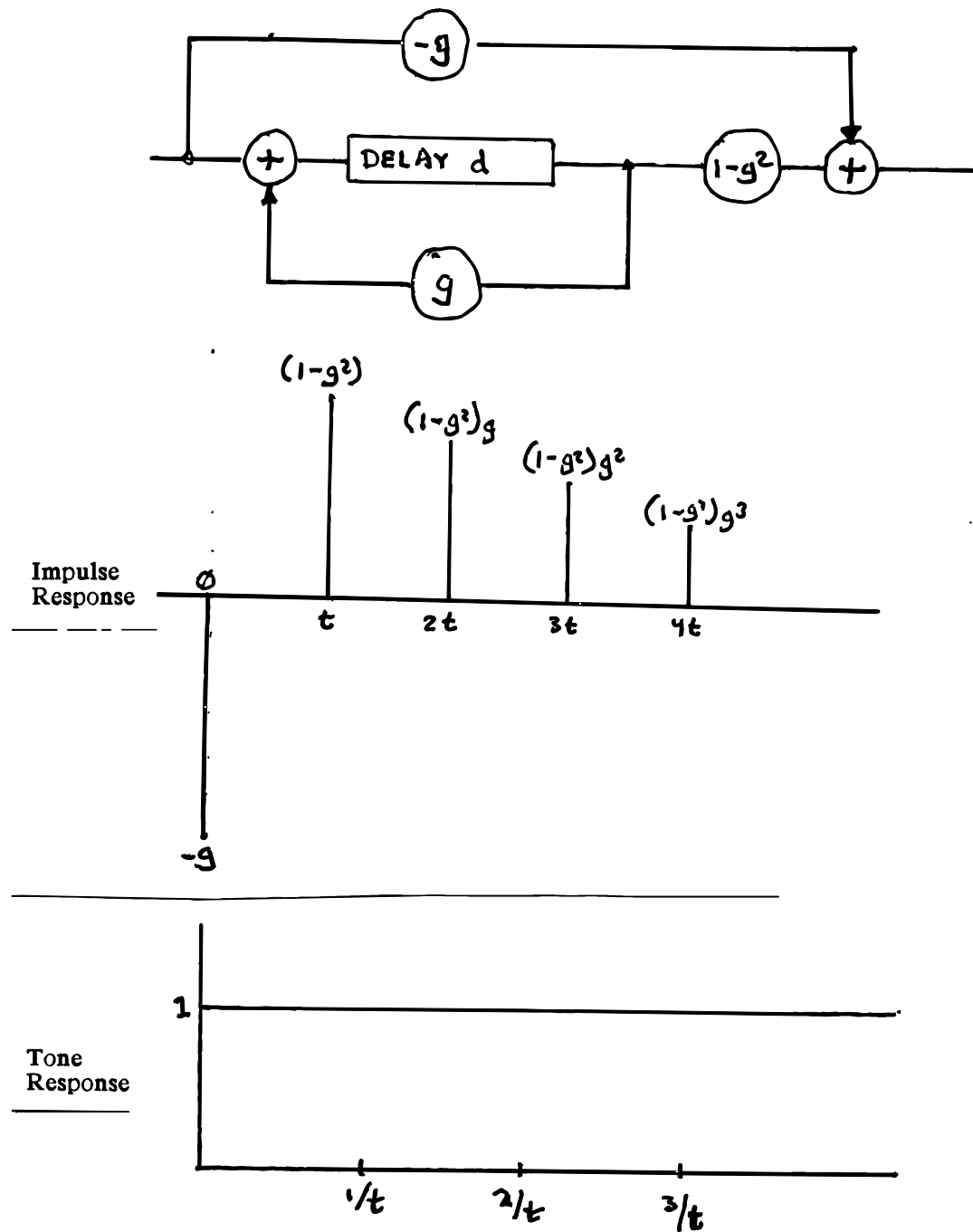


Figure 6. All-Pass network
(from Schroeder)

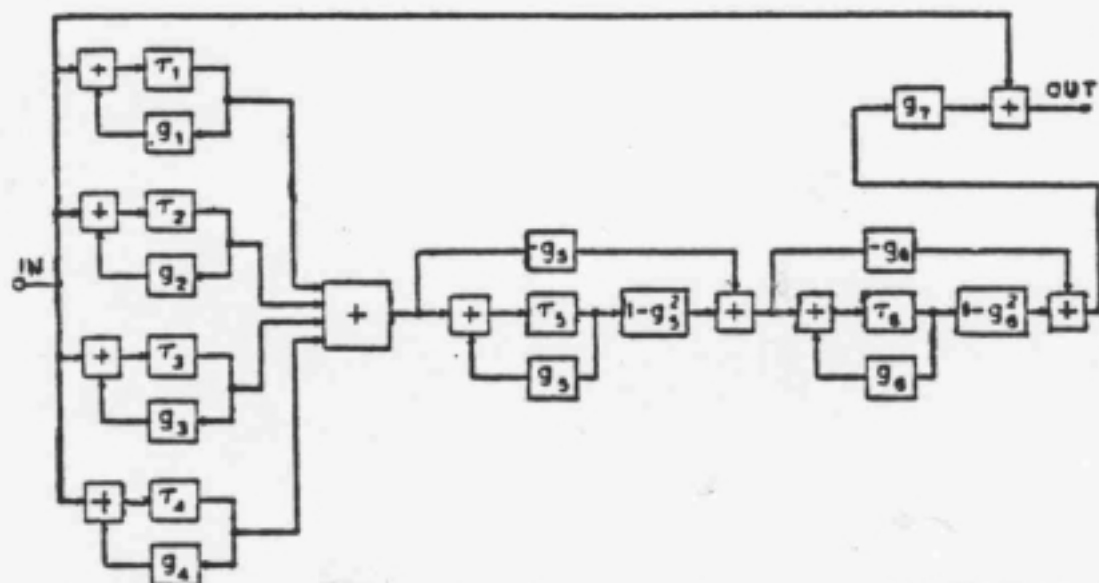


Figure 7. The Schroeder reverberator - two all-pass filters in series with 4 combs in parallel

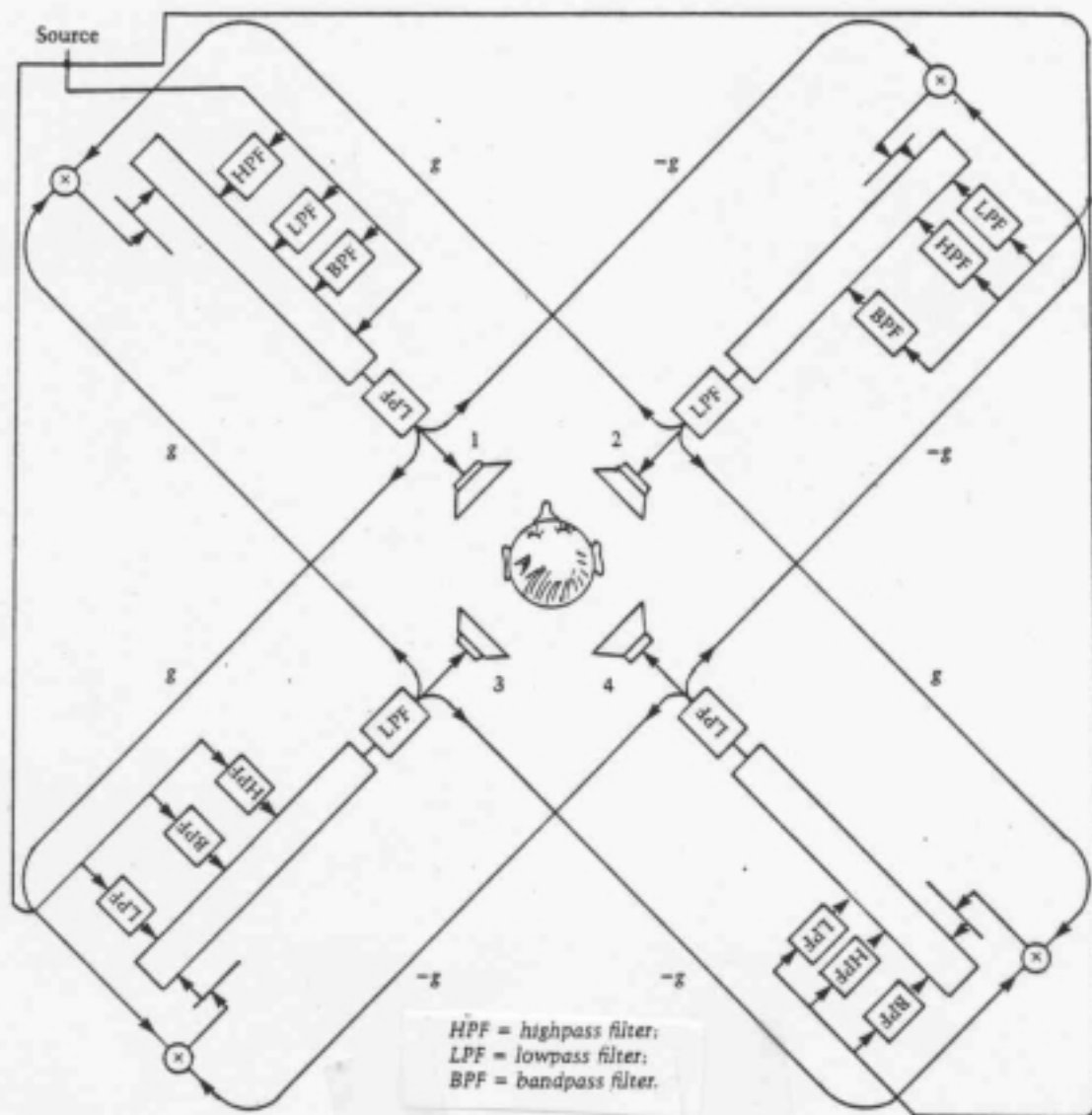


Figure 8. Stautner's reverberator - four interconnected combs